YEAR 10 PHYSICS, AQA GCSE, LESSON 5 OF 8

# PARTICLE MODEL OF MATTER

## LESSON 05

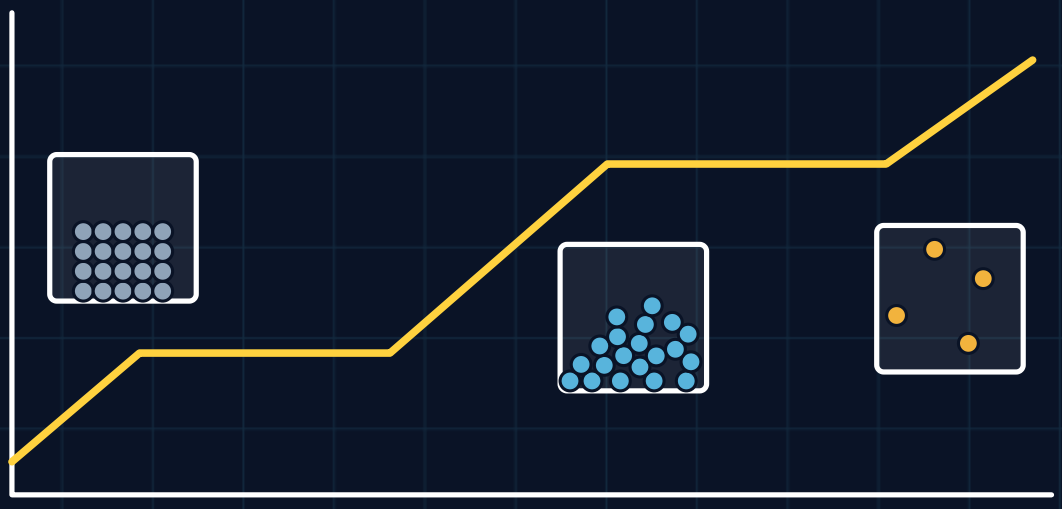
### INTERNAL ENERGY AND HEATING CURVES

A pan of water boils on the hob for ten minutes. The hob keeps supplying energy, but the water never gets hotter than 100 °C. Where does the energy go?

#### IN THIS LESSON YOU WILL

- state what internal energy is
- describe the two things that can happen when a substance is heated
- explain the flat sections of a heating or cooling graph
- read melting and boiling points from a graph

**ANSWER EDITION FOR TEACHERS AND SELF-MARKING**



NAME CLASS

START HERE

# HOW THIS BOOKLET WORKS

This is the booklet for **Lesson 5** of the Particle Model of Matter unit. It builds on the lessons before it; the "Do now" questions and the quick answers at the back help you check what you remember. Keep the booklet: it is your classwork book *and* your revision guide.

### DO NOW

Quick questions from earlier lessons. Remembering again is what makes it stick.

### LEARN IT

The explanation and diagrams. Fill in each example from the slides with your teacher, then try "Your turn" alone. Read this again when you revise.

### TRY IT

Practice that builds up: first with support, then on your own, then a challenge. Every second lesson ends with the Quick check from the slides.

### PRACTICAL

Record your measurements and work out results from them.

### EXAM CORNER

AQA-style questions with a tip on how the marks are given.

### REVISE IT

Summary, cover-and-answer questions and an "I can" check to colour in.

! **Two kinds of question to know.** When a sentence has a gap, the choices are printed straight after it, like this: The particles in a solid are close together **(close together / far apart)**. When you see a box of choices, **draw a ring around** the correct one, as AQA exam papers ask. The unit of mass is **the newton** | **the kilogram** | **the joule** .

## THE ROUTE THROUGH THE UNIT (THIS LESSON IS HIGHLIGHTED)

**01** Solids, liquids and gases

**02** Gas particles and pressure

**03** Density

**04** Measuring density (required practical)

**05** **Internal energy and heating curves**
Internal energy, the two things heating can do, heating and cooling graphs (slides 24–28)

**06** Specific latent heat

**07** Specific heat capacity

**08** Measuring specific heat capacity (required practical)

The **equation toolkit** near the back has every equation you have met so far, with formula triangles to help you rearrange. Teaching deck: **panphy.app/gcsephy/decks/particle-model**

Lesson

# 05 INTERNAL ENERGY AND HEATING CURVES

AQA Physics 4.3.2.1, 4.3.2.3; Combined Science 6.3.2.1, 6.3.2.3

**BIG QUESTION** **A pan of water boils on the hob for ten minutes. The hob keeps supplying energy, but the water never gets hotter than 100 °C. Where does the energy go?**

**BY THE END YOU CAN**

- state what internal energy is
- describe the two things that can happen when a substance is heated
- explain the flat sections of a heating or cooling graph
- read melting and boiling points from a graph

**internal energy** **kinetic energy** **potential energy** **melting point** **boiling point** **heating curve**

**YOU WILL NEED**

Per group: boiling tube of molten stearic acid (from a hot water bath), thermometer, clamp stand, stopwatch. Or use the sample results.

**Teaching deck, slides 24–28**
panphy.app/gcsephy/decks/particle-model/#24

REMEMBER

## DO NOW

From earlier lessons. No notes!

**1**
Write the equation for density.

$\rho = \frac{m}{V}$

**2**
What is the temperature of a gas related to?

The average kinetic energy of its molecules.

**3**
Name the change of state from liquid to solid.

Freezing.

EXPLAIN

## THE ENERGY INSIDE

The particles of a substance store energy in two ways. They are moving (or vibrating), so they have **kinetic energy**. They also have **potential energy**, which depends on how far apart they are: energy has to be supplied to pull particles away from each other.

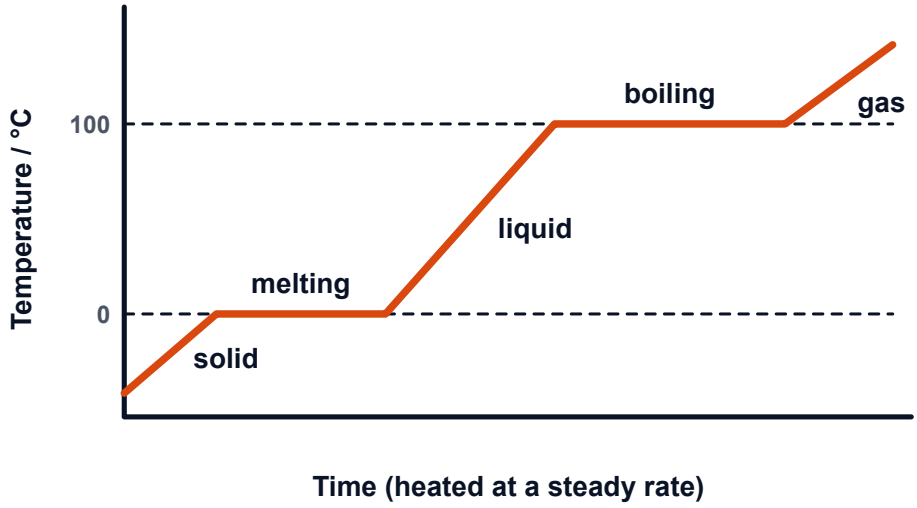
! **Internal energy** is the total kinetic energy and potential energy of all the particles (atoms and molecules) that make up a system.

## HEATING DOES ONE OF TWO THINGS

Heating a substance increases the energy of its particles, so its internal energy goes up. This **either** raises the temperature **or** changes the state. It does not do both at once.

| | THE TEMPERATURE RISES | THE STATE CHANGES |
|---|---|---|
| **What the particles do** | move or vibrate faster | break away from their neighbours and move further apart |
| **Which energy increases** | kinetic energy | potential energy |
| **Temperature** | increases | stays constant |

## A HEATING GRAPH



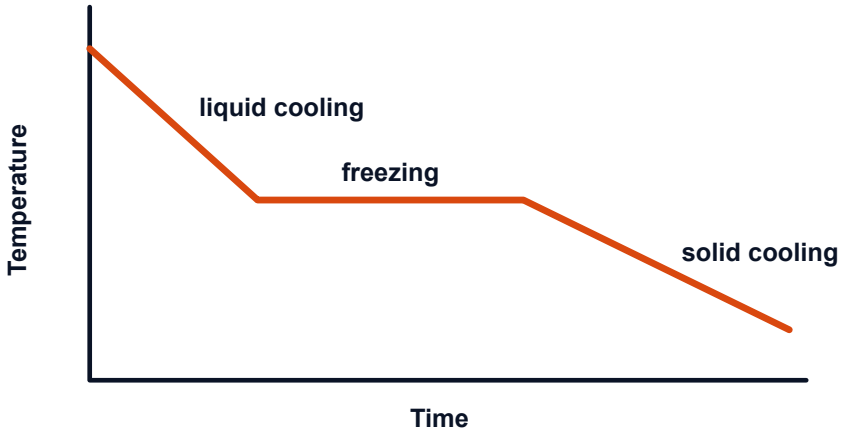

Ice from the freezer, heated at a steady rate until it is steam

- **Sloping sections:** the substance is in one state and is getting hotter. The kinetic energy of the particles is increasing.
- **Flat sections:** the substance is changing state. The energy supplied increases the potential energy of the particles, so the temperature does not change.
- The first flat section is at the **melting point** (0 °C for water). The second is at the **boiling point** (100 °C for water).

> **Get the marks:** On a flat section, heating has **not** stopped. Energy is still being supplied. It is being used to change the state, not to raise the temperature.

## A COOLING GRAPH

Cooling is the reverse. As a liquid cools, its particles slow down and the temperature falls. While it **freezes**, the particles move closer into fixed positions and their potential energy decreases: energy is transferred to the surroundings, but the temperature stays constant. Then the solid cools.




PRACTISE

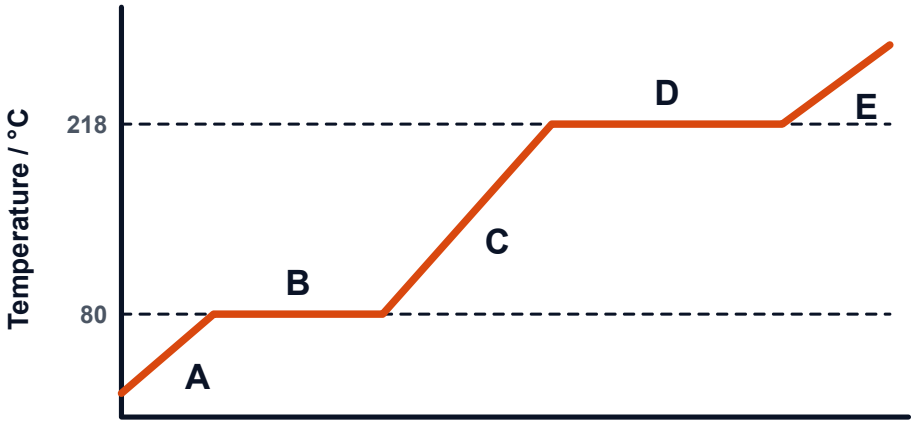
## TRY IT

**1** Complete the sentences.

Internal energy is the total ___kinetic___ **(kinetic / thermal / elastic)** energy and ___potential___ **(chemical / potential / nuclear)** energy of all the particles in a system.

While a substance is melting, its temperature ___stays the same___ **(increases / decreases / stays the same)**.

**2** A solid is heated at a steady rate until it has become a gas. The graph shows how its temperature changes.




**Time (heated at a steady rate)**

(a) Which section shows the substance melting?

| A | **B** | C | D | E |
|---|---|---|---|---|

(b) Which section shows the liquid getting hotter? | A | B | **C** | D | E |

(c) State the melting point and the boiling point of the substance.

**Melting point 80 °C; boiling point 218 °C.**

(d) In which state is the substance at 150 °C?

**Liquid.**

**3** Draw a ring around the correct answer in each box.

When water at 50 °C is heated to 60 °C, its particles gain

| **kinetic energy** | potential energy | no energy |
|---|---|---|

When water boils at 100 °C, its particles gain

| kinetic energy | **potential energy** | no energy |
|---|---|---|

**4** **CHALLENGE** 1 kg of steam at 100 °C and 1 kg of water at 100 °C are at the same temperature. Explain which has the greater internal energy.

**The steam. The particles have the same average kinetic energy (same temperature), but the particles in the steam are much further apart, so they have more potential energy.**

INVESTIGATE

## PRACTICAL: A COOLING CURVE

Hot liquid: wear eye protection and do not touch the tube.

**STAY SAFE**

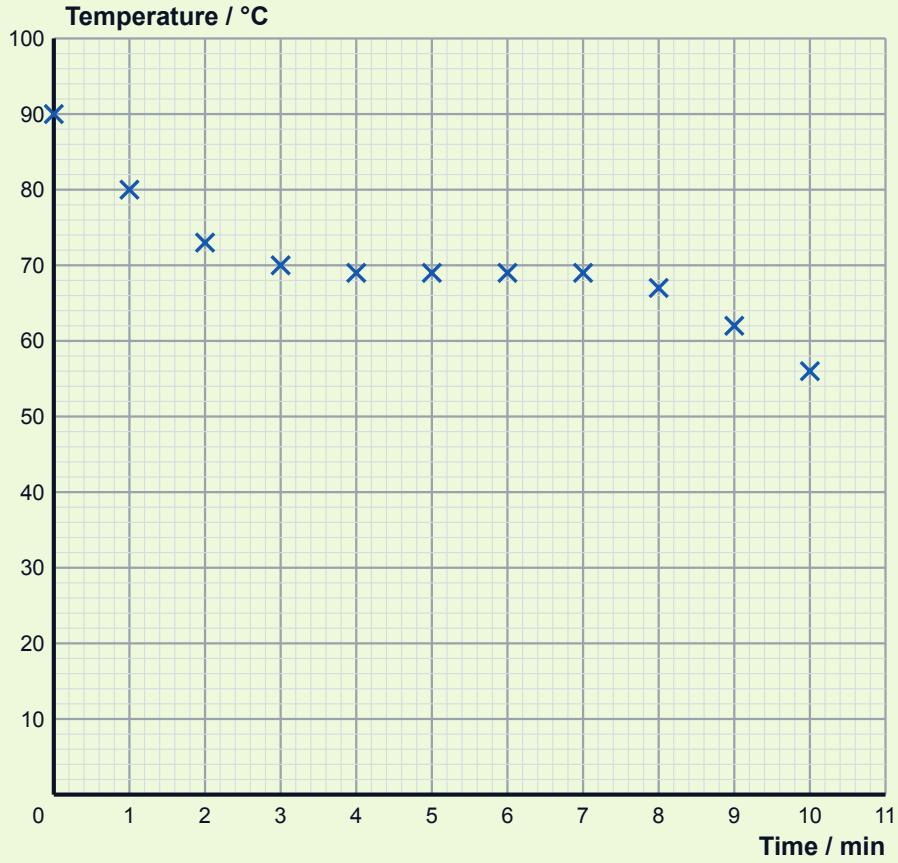
- Wear eye protection.
- The tube and the water bath are hot: leave the tube in its clamp.
- Do not stir with the thermometer once the acid starts to set: it may break.

A tube of melted stearic acid (a waxy solid at room temperature) is left to cool in air. Record its temperature every minute. If you are not doing the practical, use the sample results your teacher gives you.

**5** Record the temperature.

| Time / min | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Temperature / °C | 90 | 80 | 73 | 70 | 69 | 69 | 69 | 69 | 67 | 62 | 56 |

**6** Plot a graph of temperature against time. Draw a smooth line through your points.




**7** Use your graph to find the freezing point of stearic acid.

**About 69 °C: the temperature of the flat section.**

**8** Explain why the temperature stays the same while the stearic acid freezes, even though it is still transferring energy to the air.

**The particles are moving closer together into fixed positions, so their potential energy is decreasing. Their kinetic energy does not change, so the temperature stays constant.**

AQA-STYLE

## EXAM CORNER

Write in full sentences. Show every step of a calculation.

**Get the marks:** For a flat section, name the change of state, say that energy is still being supplied, and say which energy of the particles changes.

**9** A block of ice at 0 °C is heated. Its temperature does not change until all the ice has melted. Explain why. **[3 marks]**

**The energy supplied is used to change the state (melt the ice) (1). It increases the potential energy of the particles (as they break away from their fixed positions) (1), not their kinetic energy, so the temperature stays the same (1).**

**10** State what is meant by the internal energy of a system. **[2 marks]**

**The total kinetic energy and potential energy (1) of all the particles that make up the system (1).**

REVIEW LATER

## REVISE IT

Use this box after the lesson and before tests.

### SUMMARY

- Internal energy = total kinetic + potential energy of all the particles.
- Heating increases internal energy: the temperature rises or the state changes.
- Temperature rise: the particles gain kinetic energy.
- Change of state: the particles gain potential energy, and the temperature stays constant.
- Flat sections of a heating or cooling graph are changes of state.

### COVER THE PAGE AND ANSWER

**1** Define internal energy.

**2** What are the two things that heating a substance can do?

**3** What does a flat section on a heating graph show?

**4** Which energy of the particles increases when a substance melts?

### I CAN...

- state what internal energy is
- describe what heating does to the particles
- explain why the temperature is constant during a change of state
- interpret heating and cooling graphs

REFERENCE

# EQUATION TOOLKIT

Every equation from Lessons 1–5. AQA prints all of them on the equation sheet in the exam, but you still need to know what each symbol means, its unit, and how to use it.

1. Cover the quantity you want to find.
2. What is left side by side? **Multiply**.
3. One above the other? **Divide** the top by the bottom.

Example: to find $V$ from $\rho = \frac{m}{V}$, cover $V$. $m$ is above $\rho$, so $V = \frac{m}{\rho}$.

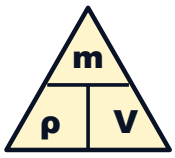
$$\rho = \frac{m}{V}$$

**density = mass ÷ volume**

kg/m³, kg, m³ (or g/cm³, g, cm³)

$m = \rho V$ and $V = \frac{m}{\rho}$

Lesson 3




**1**
Write what you know. Convert first: g → kg, cm³ → m³, kJ → J.

**2**
Write the equation in symbols.

**3**
Rearrange if you need to, then substitute the numbers.

**4**
Give the answer with a unit. Ask: is it sensible?

REFERENCE

# KEY WORDS FROM THIS LESSON

Learn the meaning and the spelling. In exams, precise words earn marks: say “the particles vibrate about fixed positions”, not “the particles shake”.

**Internal energy**: The total kinetic energy and potential energy of all the particles that make up a system

**Melting point**: The temperature at which a substance changes between solid and liquid

**Boiling point**: The temperature at which a substance boils, or condenses

CHECK YOURSELF

# QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

### DO NOW

1. $\rho = \frac{m}{V}$
2. The average kinetic energy of its molecules.
3. Freezing.

### COVER THE PAGE

1. The total kinetic and potential energy of all the particles in a system.
2. Raise its temperature or change its state.
3. A change of state (melting or boiling).
4. Potential energy.